MISO: AI Recipe Recommender & Ingredient Optimization Agent

1) Product Intro

MISO is an AI-powered recipe recommendation agent for busy people who cook at home. The app helps users answer two daily questions faster:

1. What can I cook with the ingredients I already have?
2. Which recipe should I actually make right now?

Instead of forcing users to browse hundreds of links, MISO narrows the options to a small set of highly relevant recipes from trusted content sources like YouTube, TikTok, and Instagram. The product is designed to reduce decision fatigue, save time, and reduce food waste.

2) Who It Is For

Primary users

Working professionals who come home tired, have a few ingredients in the fridge, and want a quick, tasty meal without spending 30–60 minutes searching.

Secondary users

- Students who want simple meals
- Home cooks trying to use leftovers
- Users who want healthier meals
- Users with dietary preferences such as vegetarian, keto, high-protein, or low-spice

3) Pain Points

MISO is built around three core pain points:

Decision fatigue

Users often open multiple recipe apps or videos and still cannot decide what to cook.

Food waste

Leftover ingredients often expire because users do not know how to combine them into a meal.

Low trust in search results

Generic search results are often too broad, too complex, or not aligned with the ingredients the user actually has.

4) The Solution

MISO provides a faster decision layer on top of existing recipe content.

The agent:

- understands messy user ingredient input
- finds candidate recipes from multiple sources
- removes irrelevant ingredients from consideration when needed
- ranks the best matching recipes
- learns from user likes and dislikes over time
- recommends the next best recipe if the first one is rejected

The product is not trying to generate a recipe from scratch. It is trying to select the best recipe for that user, at that moment, with the ingredients available.

5) Brief Product Sense / Execution

The product strategy is simple:

- make the first recommendation fast
- make the recommendation relevant enough to cook
- reduce unnecessary back-and-forth
- keep the user experience lightweight

The main execution decision is to optimize for **decision reduction**, not recipe exploration. That means the app should show only a few strong options, not a long list of similar links. If the agent cannot find an exact match, it should intelligently relax the constraint by selecting the best feasible ingredient combination and confirm only when needed.

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6) Core AI Product Principle

MISO is not a recipe search engine. It is an **ingredient optimization and recipe ranking system**.

The AI job is not just to search. It must:

- understand the user's pantry context
- find candidate content across platforms
- score recipe feasibility
- choose the best subset of ingredients when the full set is unrealistic
- adapt to user taste over time

This is what makes it a strong AI PM portfolio project.

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7) AI Architecture Overview

```
User Input
   ↓
Ingredient Parser / Intent Agent
   ↓
Structured User Context
   ↓
Recipe Retrieval Agent
   ↓
Candidate Recipe Pool
   ↓
Ingredient Matching + Optimization Layer
   ↓
Ranking Engine
   ↓
Top Recipe Recommendation
   ↓
Feedback Capture
   ↓
Taste Learning / Re-ranking Memory
   ↓
Next Best Recommendation
```

7.1 Ingredient Parser / Intent Agent

This layer converts natural language into structured signals.

Example input:

- spinach
- tomato
- cheese
- bread
- brinjal
- cucumber
- 15 minutes
- vegetarian
- want something tasty and easy

Parsed output:

- ingredients: spinach, tomato, cheese, bread, brinjal, cucumber
- time limit: 15 minutes
- diet: vegetarian
- preference: easy, tasty

The parser should also detect ambiguous or noisy inputs and normalize them into usable product signals.

7.2 Recipe Retrieval Agent

The retrieval layer searches for candidate recipe links across multiple sources:

- YouTube
- TikTok
- Instagram

It generates ingredient-based search queries such as:

- cheese tomato sandwich
- bread tomato cheese recipe
- easy vegetarian sandwich
- leftover veggie sandwich

The retrieval agent does not select the final recipe. It only creates a candidate pool of possible links.

7.3 Ingredient Matching / Optimization Layer

This is the most interesting part of MISO.

If the user provides a messy combination of ingredients, the agent should not force all items into one recipe. Instead, it should identify the **best recipe-compatible subset**.

Example input:

- spinach
- tomato
- cheese
- bread
- brinjal
- cucumber

The agent might decide that:

- bread
- cheese
- tomato
- cucumber work well together

And that:

- brinjal is likely irrelevant for the best available recipe

This makes MISO an optimization system rather than a rigid matching system.

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8) Matching Logic

Objective

The agent should maximize match quality, but never recommend a recipe below a minimum feasibility threshold.

Product rule

- ideal match: as high as possible
- acceptable minimum match: 75%
- below 75%: do not recommend

Match definition

The match score can be framed as:

Recipe Match Score = (available required ingredients / total required ingredients) × 100

If a recipe requires 4 ingredients and the user has 3 of them, the match score is 75%.

If the match is below threshold, the system should either:

- look for a closer recipe, or
- ask the user a single clarification question if a missing ingredient is likely available

Example behavior

User has:

- bread
- cheese
- tomato
- cucumber
- brinjal
- spinach

Best recipe may be:

- grilled cheese sandwich
- cheese tomato toast
- veggie sandwich

The system should choose the best feasible match, even if it means ignoring brinjal.

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9) Ranking Algorithm

Once candidate links are retrieved, the ranking engine chooses the best one.

Ranking goal

Select the recipe that is:

- most feasible
- most relevant
- most aligned to user taste
- easiest to execute
- most trustworthy

Suggested ranking factors

1. Ingredient Match Score — 40%

How many required ingredients does the user already have?

2. Missing Ingredient Penalty — 15%

How many extra ingredients are needed?

3. Time Fit — 10%

Does the recipe fit the user's stated time limit?

4. Taste Preference Match — 15%

Does it align with spicy, sweet, savory, healthy, cheesy, keto, or protein-rich preferences?

5. Source Quality / Content Trust — 10%

Is the source credible and the recipe likely to be usable?

6. Creator Credibility — 5%

Does the creator have strong engagement or a history of trusted recipe content?

7. Ingredient Utilization Bonus — 5%

Does the recipe use leftover ingredients that would otherwise be wasted?

Example formula

Final Score = 40% Ingredient Match + 15% Missing Ingredient Penalty + 10% Time Fit + 15% Taste Match + 10% Source Quality + 5% Creator Trust + 5% Utilization Bonus

Why this works

This ranking system helps the agent avoid recommending a viral recipe that looks good but is not actually cookable with what the user has at home.

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10) Retrieval from Content Sources

MISO should search multiple platforms because users consume recipe content differently.

YouTube

Best for detailed, step-by-step cooking videos.

TikTok

Best for short, high-engagement, quick recipe clips.

Instagram

Best for visual recipe reels and fast inspiration.

Retrieval behavior

The retrieval agent can create search queries from the parsed ingredients and preferences.

Example for bread, tomato, cheese:

- bread tomato cheese recipe
- easy sandwich recipe
- quick vegetarian snack
- cheesy toast recipe

Then the ranking engine evaluates links from all sources together.

Additional retrieval guardrail

The agent should filter out:

- broken links
- private videos
- inaccessible content
- region-restricted content
- extremely low-quality spam results

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11) Feedback Improvement / Learning System

This is the part that turns MISO into an adaptive AI product.

The system should not treat feedback as a simple thumbs-up or thumbs-down. Instead, it should learn what the user likes across multiple dimensions.

11.1 What feedback means

If the user dislikes a recipe, the system should infer whether the issue was:

- too spicy
- too sweet
- too much effort
- wrong cuisine
- too unhealthy
- too heavy
- not enough cheese
- not enough protein
- wrong creator style

11.2 Session memory vs long-term memory

Session memory

Used for immediate re-ranking in the same cooking session.

Example:

- user rejects a spicy paneer recipe
- next recommendations should reduce spicy options immediately

Long-term memory

Used to build the user's taste profile over time.

Example:

- user often likes Indian vegetarian meals
- user usually prefers quick recipes
- user saves nutritionist-led healthy content

11.3 User preference vector

The agent should maintain a hidden profile such as:

```
{
  "spicy": 0.8,
```

```
    "sweet": 0.2,
    "indian_cuisine": 0.9,
    "italian_cuisine": 0.4,
    "healthy": 0.7,
    "cheese_based": 0.6,
    "quick_meals": 0.95,
    "dietician_content": 0.8,
    "novelty": 0.3
  }
```

This profile changes based on user behavior.

11.4 Taste learning examples

If a user repeatedly likes:

- spicy recipes
- Indian recipes
- high-protein meals
- creators with nutrition backgrounds

The agent should increase the weights for those features.

If a user repeatedly dislikes:

- creamy dishes
- long recipes
- unfamiliar cuisines
- high-effort cooking videos

The agent should reduce those weights.

11.5 Next-best recommendation behavior

If the first recommendation is disliked, MISO should: 1. keep the same ingredient pool 2. infer why the recipe was rejected if possible 3. downweight the rejected attributes 4. rerank the remaining candidates 5. recommend the next best match

This makes the product feel responsive instead of static.

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12) Evaluation Framework

The agent should be evaluated across product, AI, and business metrics.

12.1 Product metrics

- time to decision
- recipe selection rate

- recipe save rate
- recipe completion rate
- repeat usage rate
- user satisfaction score

12.2 AI metrics

- ingredient parsing accuracy
- recipe match accuracy
- relevance of top recommendation
- user preference prediction accuracy
- next-best recommendation acceptance rate
- low-quality recommendation rate

12.3 Business metrics

- retention
- weekly active users
- referral rate
- monetization potential through affiliate links or creator partnerships

12.4 Core success metric

The most important metric for MISO is:

Did the user actually cook the recommended recipe?

That is the real proof that the recommendation worked.

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13) Guardrails

Because the system is making daily food decisions, it needs strong guardrails.

13.1 Match threshold guardrail

Do not recommend any recipe below 75% feasibility.

13.2 Dietary guardrail

Hard constraints like vegetarian, keto, nut allergy, or low-spice preferences should not be violated.

13.3 Safety guardrail

Never recommend unsafe cooking instructions or ingredients that conflict with food safety.

13.4 Confidence guardrail

If the system is unsure, it should either:

- ask one clarification question, or
- choose a safer fallback recipe

13.5 Excessive questioning guardrail

Do not ask too many ingredient questions. The user should not feel interrogated.

13.6 Trust guardrail

Prefer reliable creators and content that appears actionable, clear, and usable.

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14) Failure Modes

A strong AI case study should show that you understand what can go wrong.

Possible failure modes

- recommending a recipe with too many missing ingredients
- over-weighting popularity over feasibility
- ignoring dietary restrictions
- suggesting recipes that are too hard for a tired user
- learning the wrong preference from a one-time dislike
- recommending content that is unavailable or broken

How the system should respond

- rerank quickly after rejection
- use minimum match thresholds
- distinguish between short-term and long-term learning
- log failure reasons for model improvement
- apply strict filters for constraints like diet or safety

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15) MVP Scope

The first version should stay focused.

MVP includes

- ingredient input
- preference selection
- retrieval from YouTube, TikTok, and Instagram

- ranking engine
- top 3 recipe recommendations
- like/dislike feedback
- next-best recommendation

Not in MVP

- grocery delivery
- full meal planning calendar
- pantry scanning with computer vision
- voice assistant
- multi-day cooking optimization

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16) Roadmap

Phase 1: Core recommendation

Deliver fast, relevant recipe selection.

Phase 2: Adaptive feedback

Learn from likes, dislikes, saves, and cooking behavior.

Phase 3: Better optimization

Improve ingredient utilization and waste reduction.

Phase 4: Smarter personalization

Adapt more strongly to cuisine, taste, health, and creator preference.

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17) Why This Stands Out as an AI PM Portfolio Project

MISO stands out because it shows more than a basic AI wrapper.

It demonstrates:

- user problem framing
- retrieval across multiple content sources
- ingredient matching and optimization
- ranking logic with product tradeoffs
- adaptive learning from feedback
- personalization across taste dimensions

- guardrails and trust thinking
- evaluation beyond vanity metrics

This is the kind of project that shows you can think like an AI product manager, not just a user of AI tools.

Final positioning statement

MISO is an AI recipe recommendation and ingredient optimization agent that helps busy home cooks quickly decide what to make, reduce food waste, and discover recipes they are actually likely to cook. The product combines structured ingredient parsing, multi-source retrieval, ranking, personalization, feedback learning, and guardrails into one adaptive recommendation system.